Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/02/2023
PI Name:	Williams, Michael A. M.D.		
Project Title:	Regulatory Feasibility Study of Implantation of a Telemetric ICP Sensor in Commercial Spaceflight Participants		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	maw99@uw.edu	**Fax:**	FY 206-744-9942
PI Organization Type:	UNIVERSITY	**Phone:**	206-616-7291
Organization Name:	University of Washington		
PI Address 1:	Box 359924		
PI Address 2:	325 9th Avenue		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98104-2499	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	04/01/2022	**End Date:**	03/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2023 per TRISH (Ed., 7/29/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McGee, Elisa B.S. (University of Washington) Adams, Karen (University of Washington)		
Grant/Contract No.:	NNX16AO69A-EXP0010		
Performance Goal No.:			
Performance Goal Text:			
	UPDATED TASK DESCRIPTION - July 2023 This is an EXPAND Project. Spaceflight Associated Neuro-ocular Syndrome (SANS) is a disorder after long duration spaceflight characterized by swelling of the optic nerve (papilledema), folds in the choroid layer of the eye, flattening of the globe of the eye, and change in near vision. Up to 70% of astronauts develop SANS, which is considered a “red risk”, because if severe visual impairment should develop, an astronaut's ability to function safely or to control the spacecraft could be compromised. On Earth, similar findings are seen with idiopathic intracranial hypertension, a condition caused by elevated cerebrospinal fluid (CSF) pressure, also called intracranial pressure (ICP). Only a few astronauts who returned to Earth with signs of SANS have had a lumbar puncture (LP) to measure ICP. About 1/3 had elevated ICP. Non-invasive ICP methods have been used, but none have had adequate sensitivity or the capability for		

Task Description:	continuous ICP measurement or assessment of the ICP waveform and pulse pressure. The Translational Research Institute for Space Health (TRISH) convened a team of experts, including active astronauts, and leading neurologists and neurosurgeons, space physiology experts and flight surgeons with experience in commercial spaceflight to design an “ideal experiment” for measuring ICP directly before, during, and after a short orbital commercial spaceflight. Implantation of intracranial sensors was considered and rejected because of the inherent non-zero risk of brain injury. The team proposed implanting an off-the-shelf telemetric ICP monitor (Miethke M.scio) attached to a catheter in the lumbar cerebrospinal fluid (CSF) space. The Miethke M.scio and Reader Unit Set are CE approved and used clinically in Europe, but they are not U.S. Food and Drug Administration (FDA) approved. [Ed. Note: The CE marking (an acronym for the French “Conformite Europeenne”) certifies that a product has met specific European Union (EU) health, safety, and environmental requirements.] For use of the implanted M.scio sensor and Reader Unit Set in the U.S., we will need to submit an FDA Investigational Device Exemption (IDE) Pre-Submission (IDE Pre-Sub) to be followed by a full IDE submission. We seek support for the initial regulatory discovery, including the IDE Pre-Submission to be followed by a full IDE submission. Successful completion of these tasks is critical for all subsequent work on an anticipated Telemetric ICP proposal to TRISH to proceed. This study would provide NASA with important clues as to the effect of several days of zero gravity (zero G) and space environment exposure on ICP. The knowledge derived from these experiments will be used to design a safe and informative Phase 4 study which will test ICP in a long-duration spaceflight with commercial spaceflight participants or astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The successful completion of the U.S. Food and Drug Administration (FDA) QSub and Investigational Device Exemption (IDE) for early feasibility study (EFS) applications provides regulatory approval for surgical implantation of the Miethke M.scio telemetric intracranial pressure (ICP) sensor, which will demonstrate that ICP can be measured accurately and safely before, during, and after spaceflight. If the pilot early feasibility study is successful, then future research using the same FDA IDE EFS approval can be submitted to the FDA. We expect to propose future research with more participants from commercial spaceflights and from NASA, enabling ICP to be measured in a more diverse population and during longer missions. Additional physiologic measurements and experimental conditions will permit a more detailed analysis of the role of ICP in the development of Spaceflight Associated Neuro-ocular Syndrome (SANS), which is necessary to determine which countermeasures for ICP are needed for long duration spaceflight. Telemetric ICP measurements will also enable experiments to determine the effectiveness of ICP countermeasures.
Bibliography Type:	Description: (Last Updated: 05/15/2026)